

# Digital Banking App - Integration Architecture & API Contracts

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Version 1.0

## 1. System Context

NeoBank Digital connects mobile users to core banking, payment, and investment infrastructure via a microservices backend.

External Services: Onfido (KYC/identity verification), Plaid (bank account linking), FedNow/ACH (payment settlement), Visa DPS (debit card processing), Apex Clearing (brokerage), Alpaca Markets (market data), Twilio (SMS/push), AWS (infrastructure).

Internal Microservices: Auth Service (biometry, sessions), Account Service (balances, statements), Transfer Service (P2P, ACH), Card Service (virtual/physical cards), Investment Service (trading, portfolio), Notification Service (push, SMS, email), Analytics Service (spending, budgets).

## 2. Onfido KYC Integration

REQ-FIN-070: SDK integration for iOS (Swift) and Android (Kotlin) with real-time selfie liveness detection.

REQ-FIN-071: Document OCR extracts name, DOB, address, document number from ID. Results returned via webhook within 60 seconds.

REQ-FIN-072: Verification outcomes: Clear, Consider, Flagged. 'Clear' auto-provisions account. 'Consider' queues for manual review. 'Flagged' blocks account creation with reason code.

## 3. Plaid Bank Account Linking

REQ-FIN-073: Plaid Link SDK integration for external bank account connection. Supports 12,000+ financial institutions.

REQ-FIN-074: Account verification via micro-deposits (\$0.01-\$0.99) with 2-day confirmation window as fallback when instant verification unavailable.

REQ-FIN-075: Balance retrieval from linked accounts for overdraft prevention before initiating ACH transfers.

## 4. Payment Settlement

REQ-FIN-076: FedNow integration for instant P2P settlements (24/7/365). Settlement finality in < 20 seconds.

REQ-FIN-077: ACH batch processing via Nacha file format. Same-day ACH for transfers initiated before 2pm ET. Next-day for others.

REQ-FIN-078: Visa DPS (Debit Processing Service) for debit card transaction authorization, clearing, and settlement. Real-time auth response < 200ms.

## 5. Brokerage Integration

REQ-FIN-079: Apex Clearing as clearing broker. Account opening via Apex API with user's KYC data forwarded.

REQ-FIN-080: Order routing via FIX 4.2 protocol. Supports market, limit, and stop orders. Fractional shares enabled.

REQ-FIN-081: Real-time market data from Alpaca Markets API. WebSocket streaming for live price updates during market hours.

REQ-FIN-082: End-of-day portfolio valuation and unrealized P&L calculation. Settled cash vs. buying power tracking.

## 6. Sequence Flow: P2P Instant Transfer

1. Mobile App -> Auth Service: Biometric verification for transfer
2. Auth Service -> Mobile App: Session token validated
3. Mobile App -> Transfer Service: POST /api/transfers {recipient\_id, amount, memo}
4. Transfer Service -> Account Service: Validate sender balance  $\geq$  amount + hold
5. Account Service -> Transfer Service: Balance confirmed, hold placed
6. Transfer Service -> FedNow: POST /payments {sender\_routing, recipient\_routing, amount}
7. FedNow -> Transfer Service: Settlement confirmation {payment\_id, status: COMPLETED}
8. Transfer Service -> Account Service: Debit sender, credit recipient (if internal)
9. Transfer Service -> Notification Service: Notify sender (push) + recipient (push/SMS)
10. Notification Service -> Mobile App: Push notification with transfer confirmation